TECHNICAL OVERVIEW

ECAD Dyno

Closed-loop small-engine dynamometer

Electrical load control, engine instrumentation and power electronics on a custom PCB.

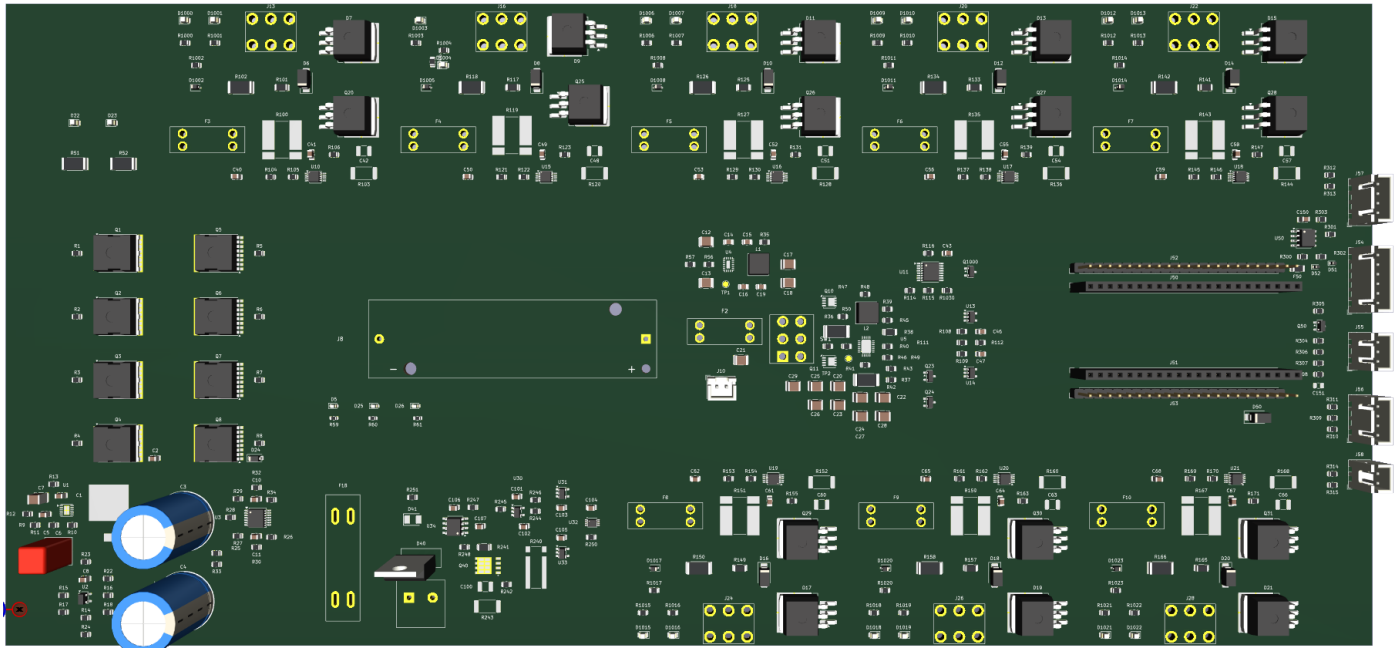


Figure 1: Preliminary KiCad component placement.

Project at a glance

ECAD Dyno is a custom electronic control and measurement platform for testing small combustion engines. An engine-driven automotive alternator provides the controllable resisting load. A separate external servo adjusts fuel delivery, while torque, rotational speed and temperature sensing provide engine feedback. The intended result is a closed-loop, repeatable engine testing platform.

Voltage	Nominal 12 V	**Controller**	ESP32-S3
Load target	150–160 A continuous	**Coarse channels**	8 high-side switches
Pulse target	Up to 200–250 A*	**Fine stage**	PWM-controlled low-side switch

*Design targets; physical testing has not yet been completed.

System concept and architecture

The system regulates the resisting torque applied by the alternator and the engine's fuel input through separate actuators. The measurement path provides feedback for subsequent control-software development, including controlled operating-point measurements and test sequences.

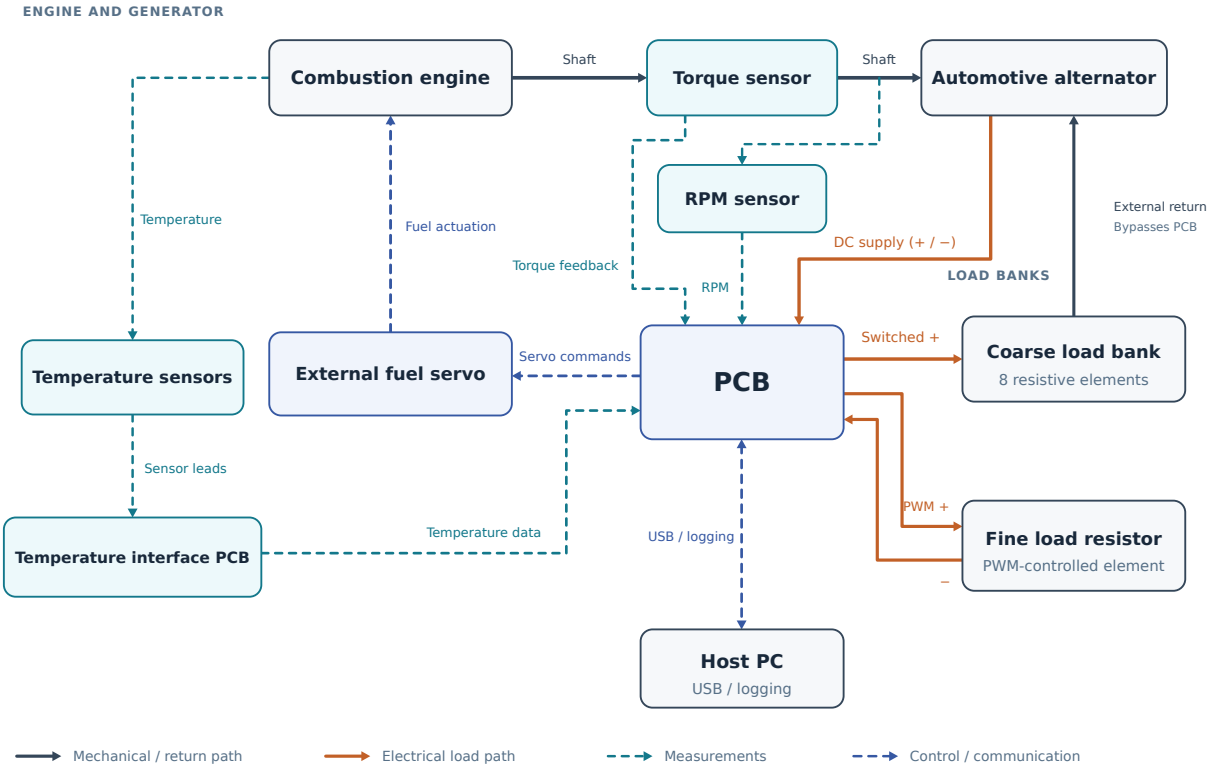


Figure 2: Dynamometer system architecture

Five electronic subsystems

Input protection	High-current input switching, voltage supervision and controlled startup.
Power management	Local DC–DC supply, $LiFePO_4$ backup and hardware power switch.
Coarse load bank	Eight resistive load channels with individual overcurrent detection and fault latching.
Fine load stage	PWM adjustment with cycle-by-cycle current limiting.
Controller and I/O	Removable ESP32-S3, engine sensors, remote peripherals and fuel-servo output.

Measurement principle

Engine shaft power is determined from measured torque and angular velocity, rather than inferred directly from alternator electrical output:

$$P_{\text{shaft}} = T\omega = \frac{2\pi n T}{60}, \qquad n \text{ in revolutions/minute.}$$

Alternator conversion and other drivetrain losses mean electrical output power is not identical to measured shaft power.

Board-level functional architecture

Five circuit sections share the protected supply, low-voltage power, control requests and hardware fault signals.

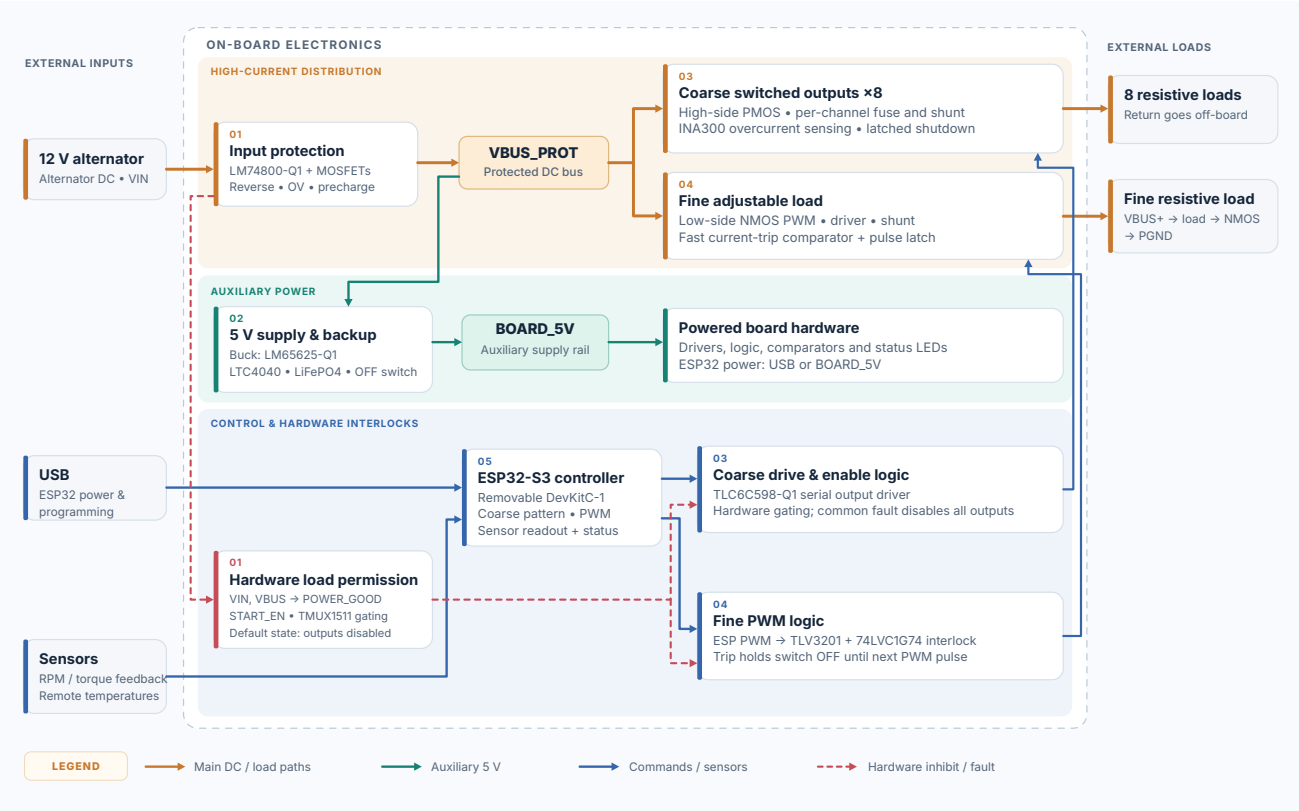


Figure 3: Board power, control and fault architecture

Separated power and control paths

- The protected high-current supply feeds the eight coarse outputs and the fine PWM load stage.
- The backed-up 5 V auxiliary rail supplies board logic and drivers; the removable controller can also be powered from USB for development.
- Hardware load-permission, channel overcurrent and fine-stage pulse limiting are intended to inhibit unsafe load commands without waiting for firmware.

High-current loading and protection

High-side coarse channels with external returns

Eight high-side MOSFET channels connect the protected positive bus to external resistive loads. A shared TLC6C598-Q1 shift register handles switching commands, with hardware fault inhibition. Normal load returns bypass the PCB.

Complete representative channel: high-side switch, shunt, INA300, local fault indication

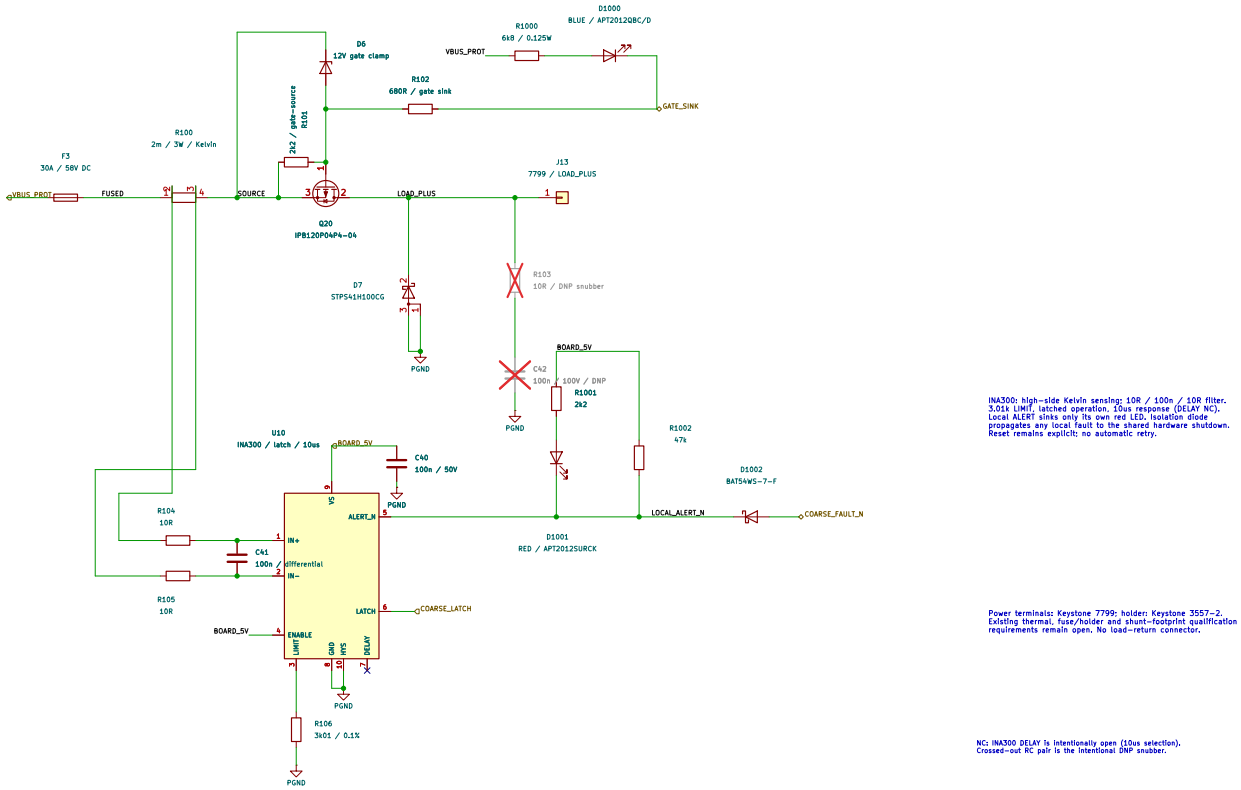
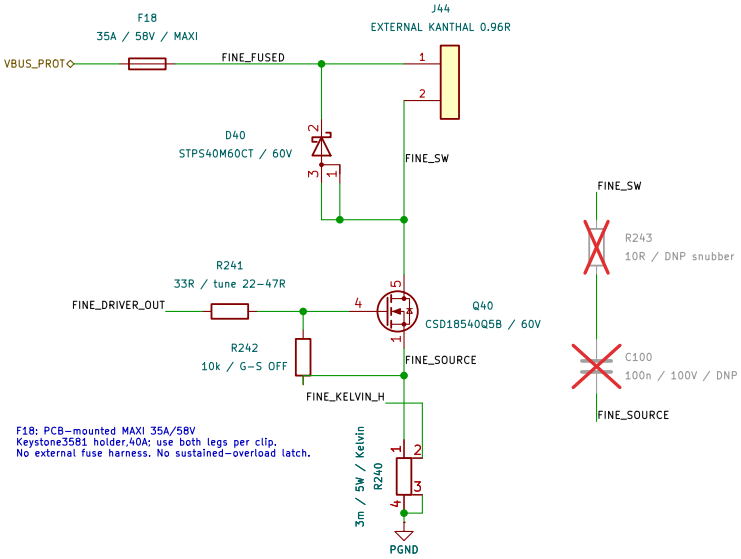


Figure 4: Coarse load channel and INA300 fault sensing

Fine PWM loading and pulse protection

A low-side MOSFET PWM stage provides finer load adjustment. The design targets 25 A continuous current, with a nominal 30.7 A static trip threshold. Hardware pulse limiting inhibits the remainder of an overcurrent PWM pulse independently of firmware.

Low-side MOSFET, resistive load and Kelvin shunt



Comparator, hardware permit and pulse memory

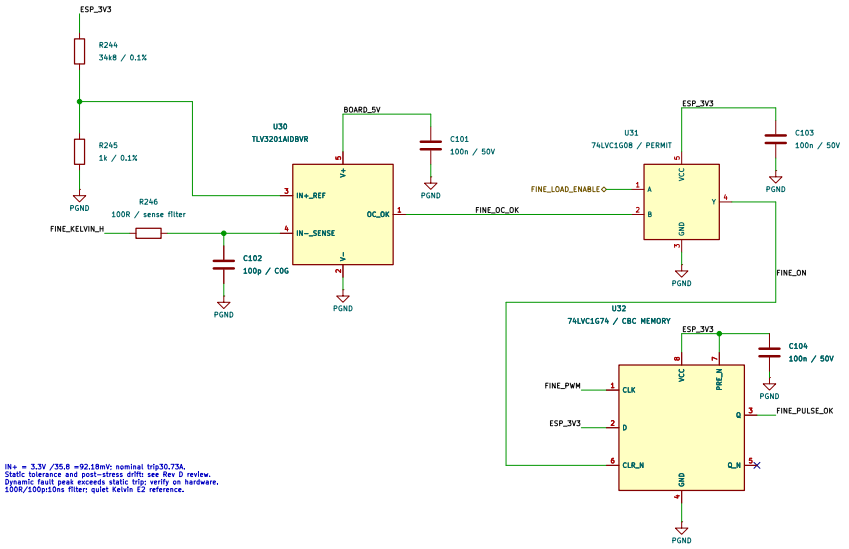
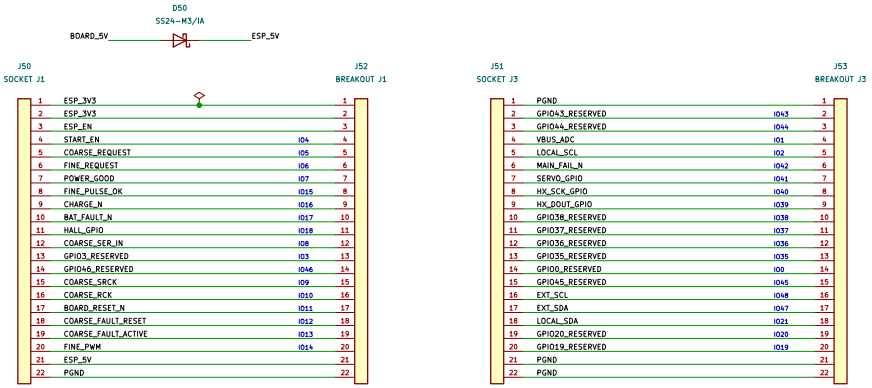


Figure 5: Fine PWM stage and pulse protection

Removable controller and external interfaces

A removable ESP32-S3-DevKitC-1 provides control and monitoring. Breakout headers expose GPIOs for diagnostics. The board interfaces with Hall/RPM sensing, an external torque module, remote temperature sensors and a fuel-control servo. A PCA9600 supports extended-distance I^{2}C.

Removable ESP32-S3 sockets and breakout headers



Extended-distance I2C interface and remote connector

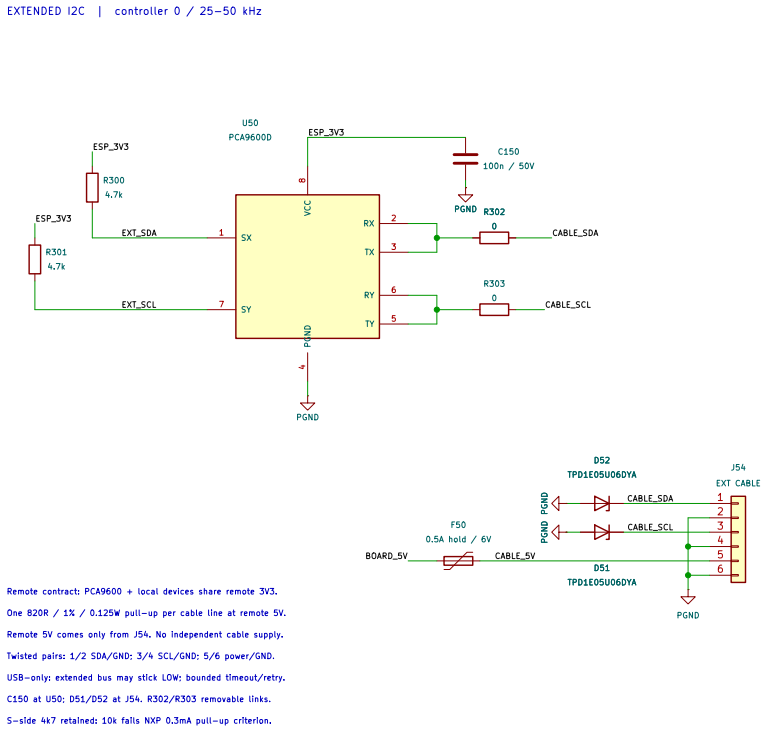
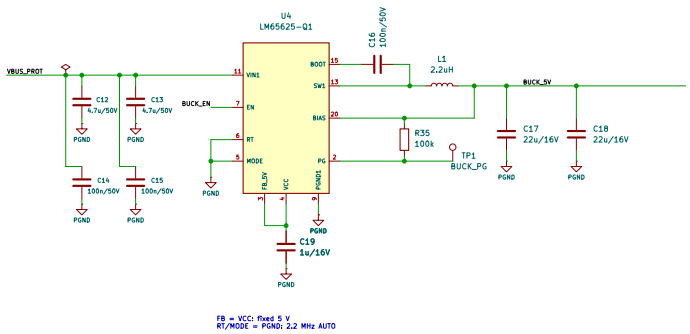


Figure 6: Removable ESP32-S3 and remote I^{2}C

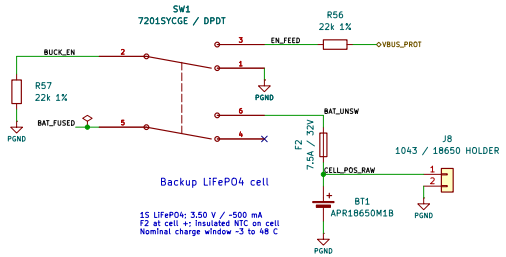
Auxiliary supply and battery backup

An LM65625-Q1 buck converter and LTC4040 backup stage supply the board electronics. A single-cell LiFePO$_4$ battery supports charging and backup operation. The hardware switch disconnects the battery and disables the buck; USB can power the ESP32 separately.

5 V buck regulator



LiFePO4 battery disconnect and cell holder



LTC4040 backup, charging and BOARD_5V path

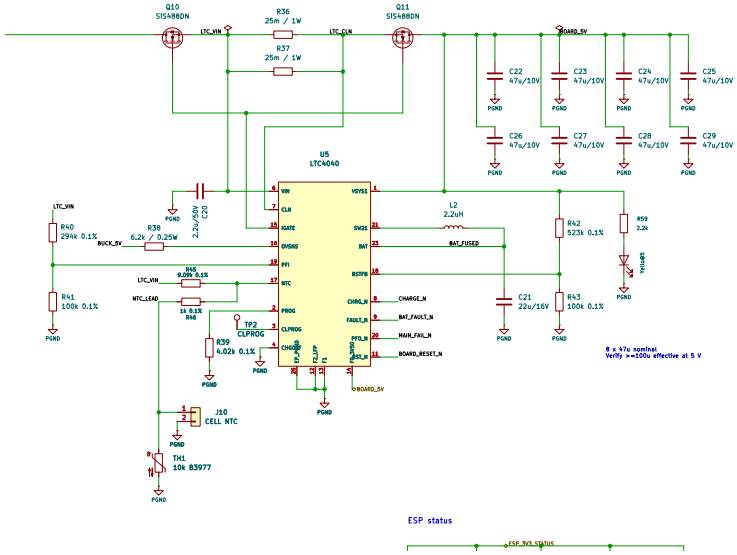


Figure 7: 5 V supply and LiFePO$_4$ backup

PCB layout and mechanical integration

Physical design considerations

High-current distribution is a central PCB design challenge. The preliminary floorplan groups coarse stages along the long board edges and incorporates a protected-positive copper busbar concept. Load switching, shunts, fuses, terminals and heat-generating devices must be arranged for short current paths, serviceable wiring and practical cooling.

- External high-current load returns, with a defined PCB power-ground strategy.
- Copper busbar integration for the protected-positive supply distribution.
- Short, low-impedance high-current interconnections and suitable device mounting.
- Thermal analysis of semiconductor devices, conductor losses and shunts.
- Accessible connectors and clearance for the removable ESP32 development board.

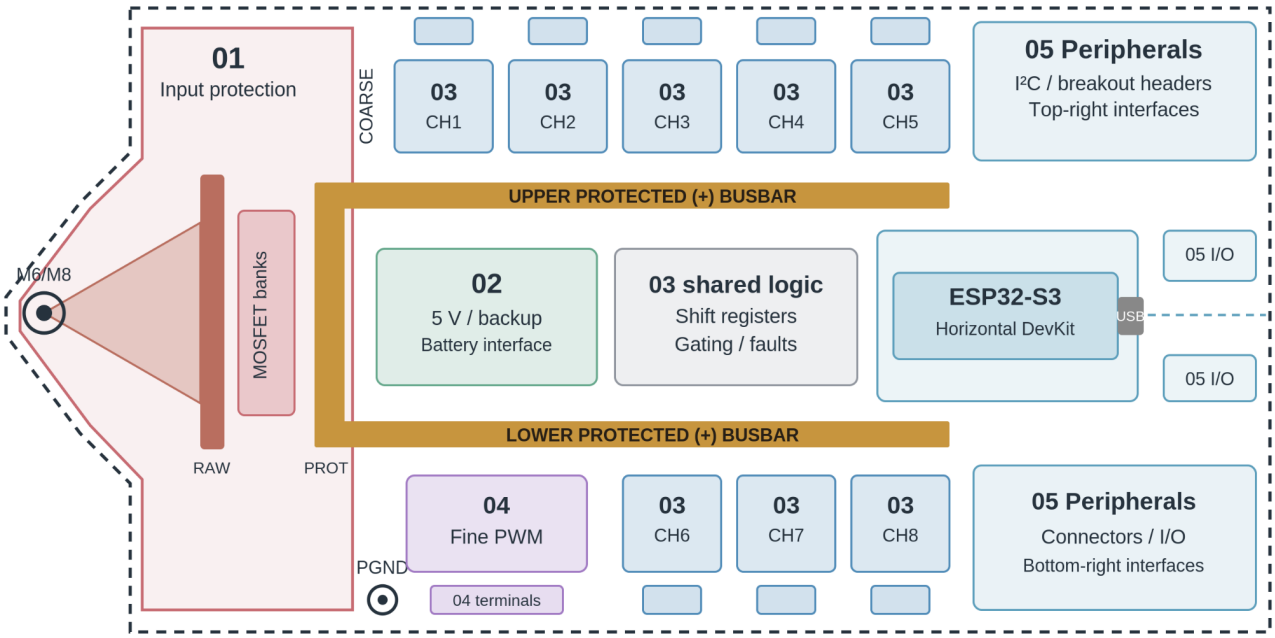


Figure 8: Preliminary PCB floorplan and busbars

Verification and development status

Design verification approach

Development proceeds from schematic design and simulation to PCB routing, fabrication and hardware testing. The current schematic passes ERC, which checks electrical-rule consistency rather than circuit performance.

Activity	Status
System architecture and circuit partitioning	Established
Hierarchical schematic design	Initial design complete; review ongoing
Schematic ERC	Passed at latest reviewed revision
Block-level circuit simulation / analytical checks	Planned
Preliminary component placement	In progress
PCB routing / clearance and thermal verification	Pending
Fabrication, assembly and electrical bench tests	Pending
Embedded firmware and closed-loop validation	Planned

Planned simulation focus

- Power-input transients, bus-capacitor charging and MOSFET switching stress.
- Regulator and backup-supply behavior in representative operating conditions.
- Comparator thresholds, gating/latched faults and fine-stage cycle-by-cycle limits.
- Worst-case conduction losses, thermal estimates and load-step conditions.

Collaboration

ECAD Dyno is being developed as a collaborative engineering project.

Input protection: switching and controlled startup

The input stage combines two series-connected MOSFET banks, surge suppression and protected-bus capacitance. An LM74800-Q1 controls switching and startup slew.

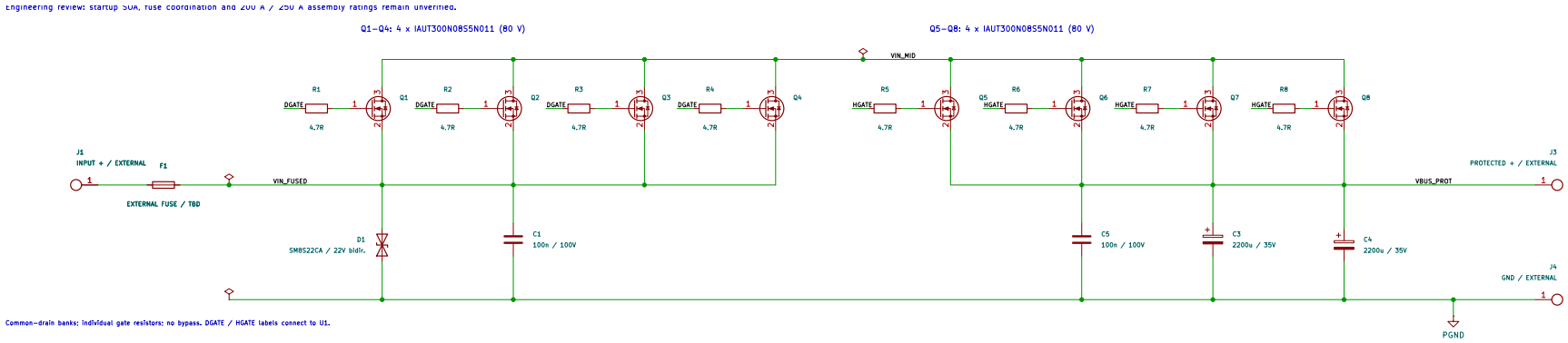


Figure 9: Input MOSFET banks and bus protection

Input protection controller and gate drive

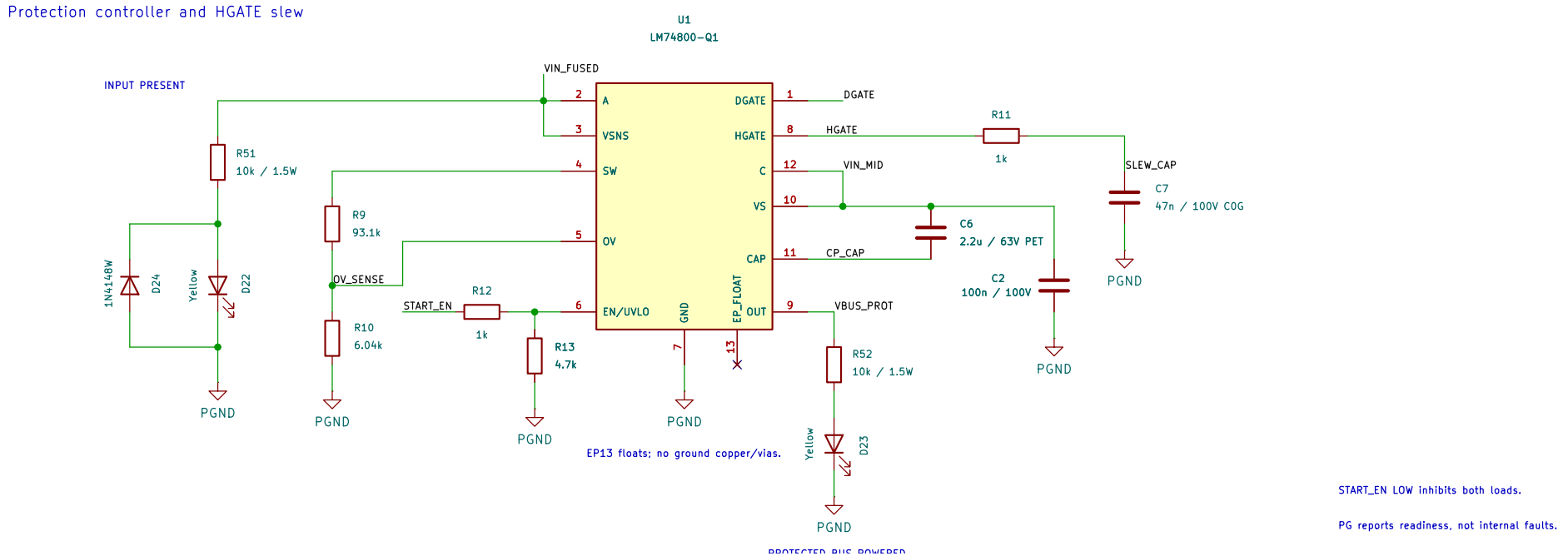


Figure 10: LM74800-Q1 protection controller

Input readiness and load gating

A comparator monitors protected-bus readiness. The TMUX1511 gates coarse and fine load-enable requests, while a voltage divider provides bus feedback to the controller.

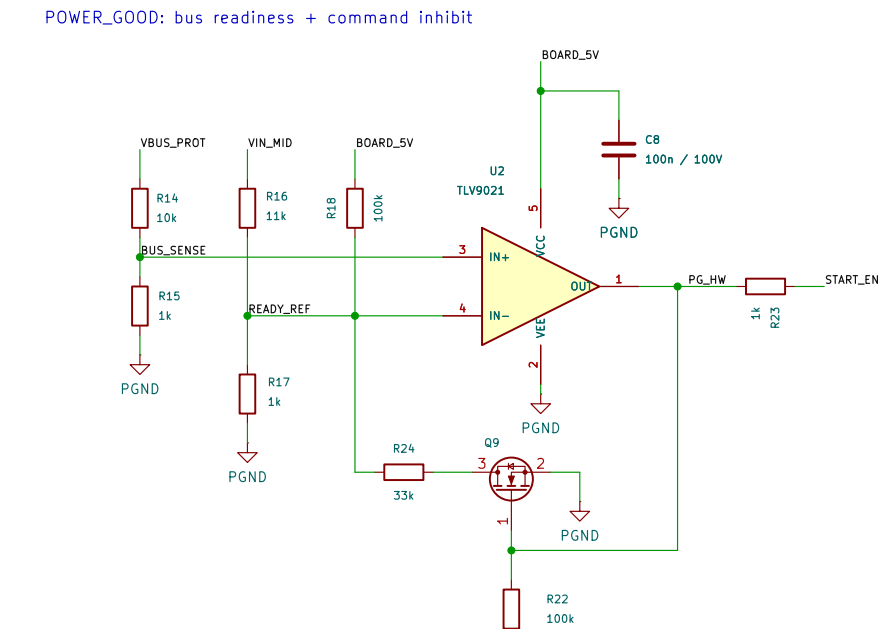


Figure 11: Protected-bus readiness comparator

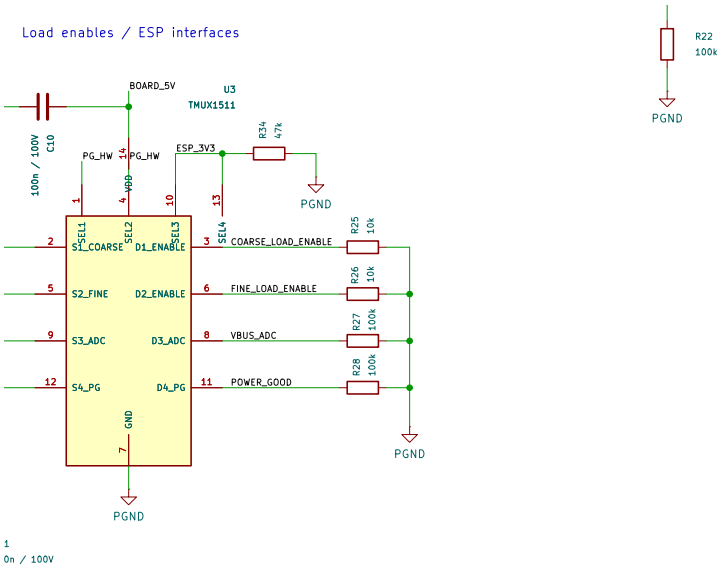


Figure 12: Load-enable gating and bus-voltage sensing

Shared coarse-channel hardware control

A shift register distributes eight channel commands. Shared hardware gating inhibits outputs on faults, with explicit reset and fault-status reporting.

Shared coarse enable, shift register, fault latch and status logic

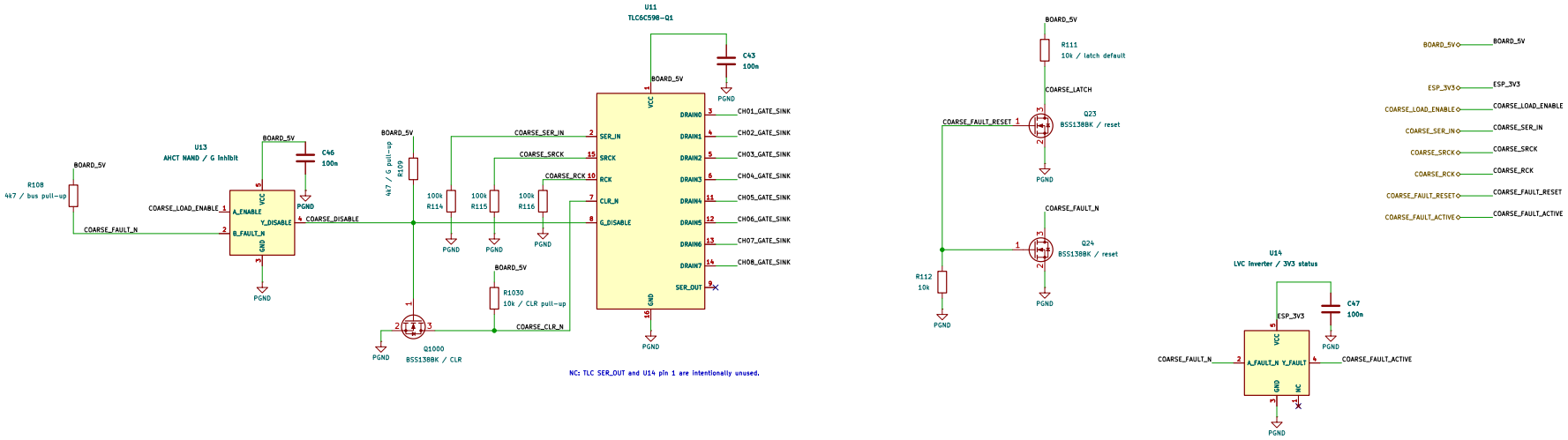


Figure 13: Coarse load commands and fault reset

Fine-stage MOSFET gate drive

The UCC27614 drives the fine-stage MOSFET through hardware-gated PWM. Local passives establish OFF defaults and complete the gate-drive path.

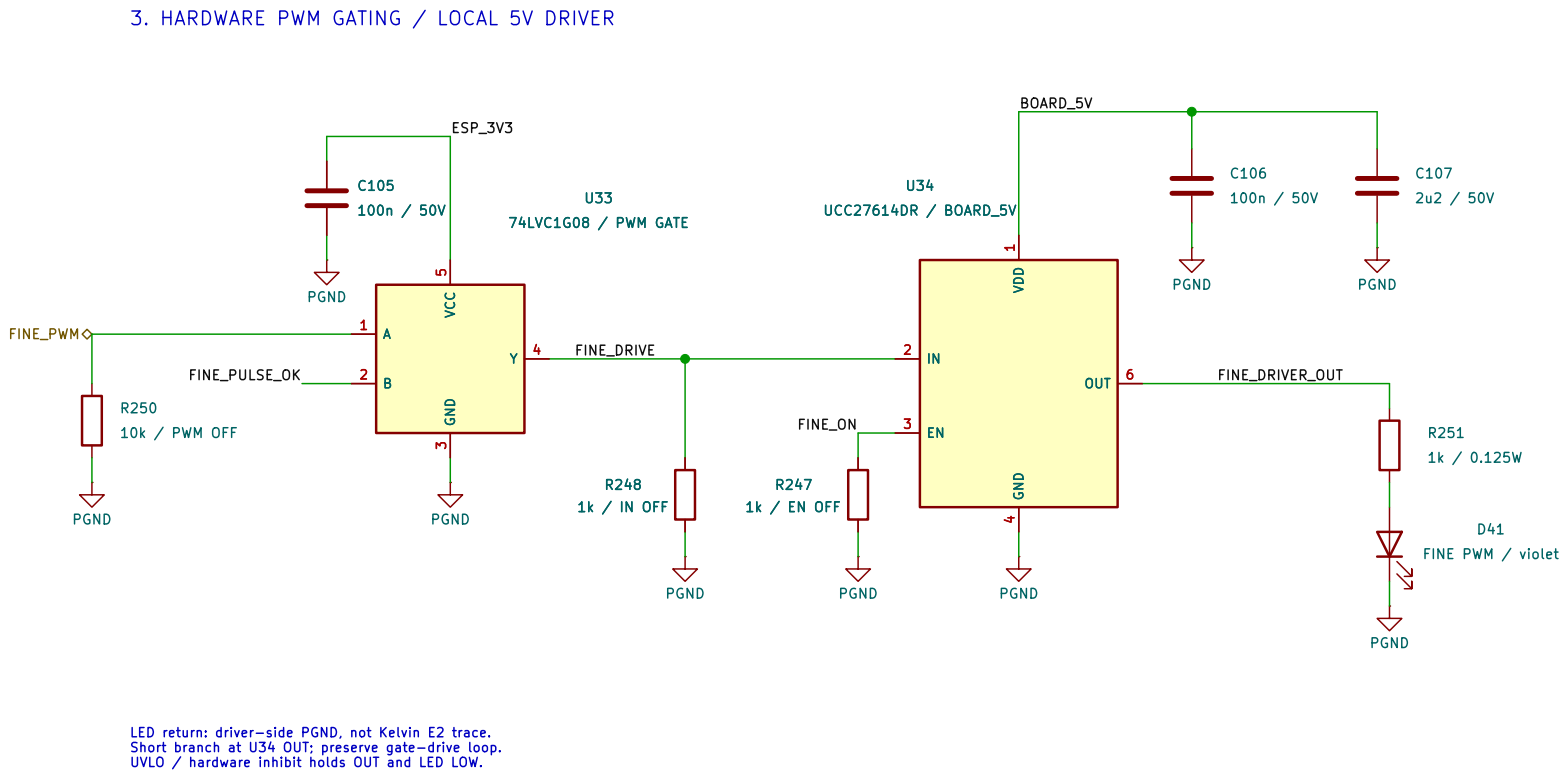
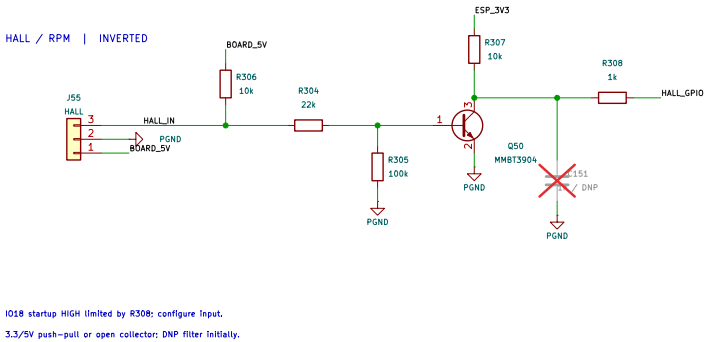


Figure 14: Fine-stage PWM gate driver

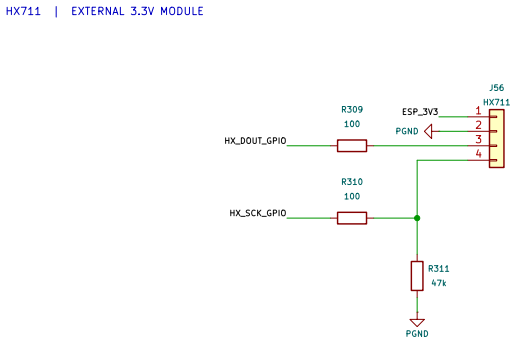
Engine instrumentation and servo connections

Sensor connections include Hall/RPM input conditioning, an external HX711 torque interface, local I^{2}C and a signal-only servo output.

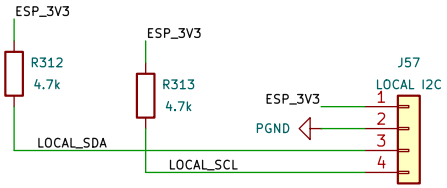
Hall sensor / engine RPM



External HX711 torque-sensor module



Local I2C sensor interface



External fuel-servo PWM signal

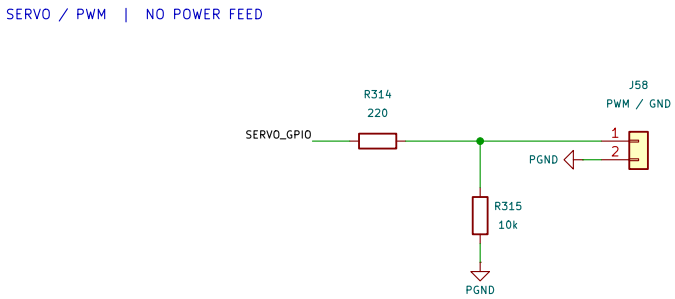


Figure 15: Engine sensor and servo interfaces